RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Sunday, 30 August 2026, 10:43 PM
State	Finished
Completed on	Sunday, 30 August 2026, 10:46 PM
Time taken	3 mins 18 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int cmpAsc(const void *a, const void *b) {
    int x = *(int *)a;
    int y = *(int *)b;
    if (x < y) return -1;
    if (x > y) return 1;
    return 0;
}

int cmpDesc(const void *a, const void *b) {
    int x = *(int *)a;
    int y = *(int *)b;
    if (x < y) return 1;
    if (x > y) return -1;
    return 0;
}

int main() {
    int n;
    scanf("%d", &n);

    if (n <= 0) {
        printf("0\n");
        return 0;
    }

    int *A = (int *)malloc(n * sizeof(int));
    int *B = (int *)malloc(n * sizeof(int));

    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &B[i]);
    }

    // Sort A ascending, B descending
    qsort(A, n, sizeof(int), cmpAsc);
    qsort(B, n, sizeof(int), cmpDesc);

    long long minSum = 0;
    for (int i = 0; i < n; i++) {
        minSum += (long long)A[i] * B[i];
    }

    printf("%lld\n", minSum);

    free(A);
    free(B);
    return 0;
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course